DynamoDB tables

Table names, partition keys (PK), sort keys (SK) and indexes for the Muster ERP backend.

Summary

Table	PK	SK	Indexes (name: PK / SK)
erp_organizations	orgId	none	none
erp_employees	orgId	empId	**ByManager**: managerId / empId **ByEmail**: email / none
erp_attendance	empId	date	**ByOrgDate**: orgDate / empId
erp_leave_balances	empId	typeId	none
erp_leave_requests	empId	reqId	none
erp_expenses	empId	reqId	none
erp_approvals	reqId	none	**ByApprover**: approverKey / createdAt **ByOrgStatus**: orgStatus / createdAt
erp_assets	orgId	assetId	none
erp_payroll_runs	orgId	period	none
erp_payroll_lines	runId	empId	**ByEmployee**: empId / period
erp_documents	empId	documentId	none
erp_notifications	empId	notificationId	none
erp_audit_logs	orgId	auditId	none

All keys are strings. Every table uses on-demand billing. Indexes are global secondary indexes that project all attributes. Table names start with the prefix in `TABLE_PREFIX` (default `erp`). The definitions live in `config/tables.js`, and `npm run db:create` creates them.

Table details

erp_organizations

One record per company. Holds the shift and attendance rules, plus the small lists of departments and leave types.

Key	Attribute	Example value
PK (partition key)	**orgId**	kestrel
SK (sort key)	**none**	

No indexes.

Other attributes: name, timezone, shiftName, shiftStart, shiftEnd, graceMinutes, halfDayAfterMinutes, geofence, departments (list), leaveTypes (list)

erp_employees

Every person, including their sign-in details. Bank account and national ID are stored encrypted.

Key	Attribute	Example value
PK (partition key)	**orgId**	kestrel
SK (sort key)	**empId**	emp_sanjay

Index name	Index PK	Index SK	Used for
ByManager	managerId	empId	A manager's direct reports. Only employees that have a managerId appear.
ByEmail	email	none	Sign in: find the employee by email.

Other attributes: empCode, name, email, phone, role, deptId, designation, managerId, joinDate, status, salary, passwordHash, bankAccountEnc, nationalIdEnc

erp_attendance

One record per person per day. A second check-in on the same day is rejected because the key already exists.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**date**	2026-10-13

Index name	Index PK	Index SK	Used for
ByOrgDate	orgDate	empId	Everyone in a company on one day (team and company attendance, dashboard, trend).

Other attributes: orgId, orgDate (orgId#YYYY-MM-DD), deptId, status, checkInAt, checkOutAt, lateByMinutes, workedMinutes, lat, lng, deviceId

erp_leave_balances

How many days of each leave type a person has, has used, and has waiting for approval.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**typeId**	lt_casual

No indexes.

Other attributes: name, color, total, used, pending

erp_leave_requests

A person's leave requests. reqId is the same id as the matching record in approvals.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**reqId**	lr_seed01

No indexes.

Other attributes: orgId, typeId, typeName, from, to, days, reason, status, createdAt

erp_expenses

A person's expense claims. reqId is the same id as the matching record in approvals.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**reqId**	lr_seed01

No indexes.

Other attributes: orgId, category, amount, date, description, status, createdAt

erp_approvals

One record per request that needs a decision (leave or expense). The decision history is kept inside it in the actions list.

Key	Attribute	Example value
PK (partition key)	**reqId**	lr_seed01
SK (sort key)	**none**	

Index name	Index PK	Index SK	Used for
ByApprover	approverKey	createdAt	An approver's inbox. approverKey is the manager's empId, or orgId#ROLE. Removed when the request closes.
ByOrgStatus	orgStatus	createdAt	Company requests by status (orgStatus = orgId#PENDING, orgId#APPROVED ...).

Other attributes: orgId, entityType (LEAVE or EXPENSE), requesterId, requesterName, requesterManagerId, summary, payload, stages, totalStages, currentStage, approver, approverKey, status, orgStatus (orgId#STATUS), actions (list), createdAt, completedAt

erp_assets

Company equipment. Every assign and return is added to the history list inside the record.

Key	Attribute	Example value
PK (partition key)	**orgId**	kestrel
SK (sort key)	**assetId**	as_laptop1

No indexes.

Other attributes: tag, name, category, serial, status, assignedTo, assignedToName, assignedAt, value, purchaseDate, history (list)

erp_payroll_runs

One payroll run per month. Status moves DRAFT, then PROCESSING (calculated), then FINALIZED (locked).

Key	Attribute	Example value
PK (partition key)	**orgId**	kestrel
SK (sort key)	**period**	2026-09

No indexes.

Other attributes: status, employeeCount, totalGross, totalNet, createdBy, finalizedBy, finalizedAt

erp_payroll_lines

One pay line per person per run. runId is orgId#YYYY-MM.

Key	Attribute	Example value
PK (partition key)	**runId**	kestrel#2026-09
SK (sort key)	**empId**	emp_sanjay

Index name	Index PK	Index SK	Used for
ByEmployee	empId	period	A person's own payslips across months.

Other attributes: orgId, period, empCode, name, designation, basic, hra, allowances, gross, workingDays, lopDays, lopAmount, pf, professionalTax, tds, deductions, net

erp_documents

Details of files stored in S3, owned by one person. The file itself is in S3, key below.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**documentId**	doc_1a2b3c4d5e

No indexes.

Other attributes: orgId, category, filename, key (S3 key), size, contentType, uploadedBy, uploadedAt

erp_notifications

In-app notifications for a person. notificationId is the time plus a random suffix, so the newest sort last.

Key	Attribute	Example value
PK (partition key)	**empId**	emp_sanjay
SK (sort key)	**notificationId**	2026-10-13T08:00:00.000Z#n_ab12

No indexes.

Other attributes: title, body, link, read, createdAt

erp_audit_logs

Append-only record of sensitive actions. auditId is the time plus a random suffix.

Key	Attribute	Example value
PK (partition key)	**orgId**	kestrel
SK (sort key)	**auditId**	2026-10-13T08:00:00.000Z#a_cd34

No indexes.

Other attributes: at, actorId, actorName, actorRole, action, entityType, entityId

Which query uses which key

What the app needs	Table	Index	Looked up by
Sign in	`employees`	`ByEmail`	email
Open one person	`employees`	-	orgId + empId
List the company's people	`employees`	-	orgId
A manager's team	`employees`	`ByManager`	managerId
Today's check-in status	`attendance`	-	empId + date
My attendance for a month	`attendance`	-	empId, date between month start and end
Everyone on a given day	`attendance`	`ByOrgDate`	orgId#date
My leave balances / requests	`leaveBalances` / `leaveRequests`	-	empId
My expenses	`expenses`	-	empId
Open one request	`approvals`	-	reqId
My approval inbox	`approvals`	`ByApprover`	empId, and orgId#ROLE
Requests by status (HR history)	`approvals`	`ByOrgStatus`	orgId#STATUS
Company assets	`assets`	-	orgId
Payroll months	`payrollRuns`	-	orgId
Pay lines of a run	`payrollLines`	-	orgId#period
My payslips	`payrollLines`	`ByEmployee`	empId
My documents	`documents`	-	empId
My latest notifications	`notifications`	-	empId, newest first
Recent audit entries	`auditLogs`	-	orgId, newest first

S3 bucket

Documents are stored privately in the bucket named by `S3_BUCKET`, and only ever opened through short-lived links.

Object key
`{orgId}/{empId}/{category}/{documentId}-{filename}`
Example: kestrel/emp_sanjay/identity/doc_1a2b3c4d5e-passport.pdf

Categories: identity, bank, offer-letters, payslips, certificates, contracts, assets. The matching record is in `erp_documents`.